Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-18-08-C01				
NPLEX-P1-CARD-18-08-C02				
NPLEX-P1-CARD-18-08-C03				
NPLEX-P1-CARD-18-08-C04				
NPLEX-P1-CARD-18-08-C05				
NPLEX-P1-CARD-18-08-C06				
NPLEX-P1-CARD-18-08-C07				
NPLEX-P1-CARD-18-08-C08				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C01-BD1

Build the Virchow triad and the two thrombus types

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C01 Activity: NPLEX-P1-CARD-18-08-C01-BD1 Date: _____

Without looking at your notes, write out the three limbs of the Virchow triad. Under each limb write the mechanism in one or two sentences and list the clinical settings that act through it. Then draw an artery and a vein side by side and build a thrombus in each. Beside each one write its color, what it is mostly made of, how firmly it attaches to the wall, the direction it grows relative to blood flow, and the vessels or chambers where it typically forms. Finish by listing the ways normal endothelium keeps blood fluid, naming the molecules involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C01-BD2

Tell a thrombus formed in life from one formed after death

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C01 Activity: NPLEX-P1-CARD-18-08-C01-BD2 Date: _____

A pathologist removes two red masses at autopsy. One lifts out of the femoral vein in one piece, is gelatinous, dark red below and yellow above, and shows no layering. The other is attached to the wall of the left atrium and shows alternating pale and dark bands. For each mass write when it formed, the features that prove it, and why the layering appears in one and not the other. Then write which of the two could have produced the pulmonary embolus found in the lung, and explain the physical reason the bands form only in moving blood. Finish by listing the five fates open to a thrombus formed in life, with the consequence of each, and write which fate explains a fibrous thickening in a vein wall found years later.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C02-BD1

The brakes on thrombin and the inherited thrombophilias

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C02 Activity: NPLEX-P1-CARD-18-08-C02-BD1 Date: _____

Draw thrombin in the centre of the page and write around it everything thrombin does to fibrinogen, to platelets, and to other clotting factors. Then draw the three brakes that restrain it: write the name of each brake, the factors it inactivates, and what speeds or assists it. Underneath, build a four row table of the inherited hypercoagulable states with columns for the defect, the mechanism of thrombosis, and one distinguishing note for each. Finish by writing whether these states mainly cause arterial or venous thrombosis and listing the four clinical clues that should make you suspect an inherited thrombophilia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C02-BD2

Three clotting puzzles explained from the mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C02 Activity: NPLEX-P1-CARD-18-08-C02-BD2 Date: _____

Work each of these from the mechanism. First, a woman has a prolonged partial thromboplastin time in the laboratory and a deep vein thrombosis in her leg, and her nontreponemal syphilis test is positive although she has no syphilis. Name the disorder, the antibody target, and explain how one disorder produces a long clotting time in the tube and clotting in the patient. Second, a man with low protein C begins warfarin and within days develops patches of skin necrosis. Write which proteins warfarin lowers, the order in which they fall, and why the first days of treatment make him more likely to clot rather than less. Third, a woman has her first deep vein thrombosis at 26 weeks of pregnancy and is found to carry factor V Leiden. Write what pregnancy contributes through each limb of the Virchow triad and why the inherited trait had caused no thrombosis before. Finish with one sentence on why heparin is less effective in antithrombin deficiency.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C03-BD1

Where an embolus lodges, and the nonthrombotic emboli

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-BD1 Date: _____

Draw the circulation in outline with the right heart, the lungs, the left heart, and the systemic arteries. On the drawing trace where material entering a systemic vein ends up, where material from the left heart or aorta ends up, and the one route by which a venous embolus reaches a systemic artery. Name that route and the defects that provide it. Then write the sources of systemic thromboemboli with the approximate share each contributes, and the organs where they lodge with the approximate share each receives. Underneath, build a table of the nonthrombotic emboli with one row each for fat and marrow, amniotic fluid, air, nitrogen, and atheroembolism, and columns for the setting, the mechanism, and the findings.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C03-BD2

Three pulmonary emboli of different size

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-BD2 Date: _____

Three patients each have a pulmonary embolus. The first is small and in a peripheral branch, the second is medium sized and the patient has long standing left heart failure, and the third sits astride the bifurcation of the main pulmonary artery. For each patient write what happens to the lung tissue, whether an infarct forms and why, and what happens to the right ventricle and the circulation. Then explain from ventilation and perfusion why the second patient is hypoxemic, what happens to dead space, and why the arterial carbon dioxide falls rather than rises. Finish by writing what repeated small emboli do to the pulmonary circulation over months and why more than 95% of these emboli come from the legs above the knee rather than the calf.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C03-RT01

Rebuild the route from a leg vein to a blocked pulmonary artery

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C03 Activity: NPLEX-P1-CARD-18-08-C03-RT01 Date: _____

Rebuild the six step sequence that runs from conditions in a deep leg vein to a blocked pulmonary artery, one step per line and in order. Beside the first step name which limb or limbs of the Virchow triad are usually acting. Beside the step where the fragment detaches, write which part of the thrombus breaks away and why that part is loosely held. Beside the final step write what is now true of ventilation and perfusion in that part of the lung and which one of the two is preserved. Then draw the chambers the fragment passes through, in order, and mark the point at which a right to left shunt would send it somewhere else instead.

Rebuild from memory: From deep vein thrombosis to pulmonary embolism

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C04-BD1

Draw an infarct and date it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C04 Activity: NPLEX-P1-CARD-18-08-C04-BD1 Date: _____

Draw a wedge shaped infarct in a kidney. Mark the occluded vessel, the apex, the base, the organ surface, and the margin, and write what covers the overlying serosa and why. Then write the two separate questions that describe every infarct, and for each question give the two possible answers. Underneath, write out the time course of coagulative necrosis from the moment of occlusion: when cell death first becomes visible under the light microscope, when nuclei disappear and the cytoplasm turns eosinophilic, when neutrophils reach the border, when macrophages and granulation tissue appear, and what the area becomes at the end. Finish by writing what the brain does instead at each of those stages and what it leaves behind.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C04-BD2

Same occlusion, four different outcomes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C04 Activity: NPLEX-P1-CARD-18-08-C04-BD2 Date: _____

The same sized embolus lodges in four patients. Patient one has it in a renal artery branch. Patient two has it in a pulmonary artery branch and a normal heart. Patient three has it in a middle cerebral artery branch and the embolus breaks up and flow returns after a few hours. Patient four has it in a coronary artery that has been narrowing slowly for ten years. For each patient write whether an infarct forms, its color, the necrosis pattern, and what the area looks like weeks later, and explain each answer from the blood supply of the organ and the tissue itself. Then write which single patient would have a larger infarct if his hemoglobin were 7 g/dL and explain why. Finish by writing the four factors that decide whether and how large an infarct forms, with one line on each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-08-C04-CP01

White against red infarcts

Fill in every cell of the table from memory, then check it against the notes and correct what you missed.

Feature	White (pale, anemic) infarct	Red (hemorrhagic) infarct
Type of vascular occlusion behind it		
Kind of tissue it occurs in		
Why blood is absent or present in the dead area		
Organs and situations where you see it		
Usual necrosis pattern in that organ		
What the area looks like weeks later		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C05-BD1

Patterns of bleeding into tissue

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C05 Activity: NPLEX-P1-CARD-18-08-C05-BD1 Date: _____

Build a four row table of hemorrhage into tissues with one row each for petechiae, purpura, ecchymosis, and hematoma. For each row write the size and form and the typical causes. Then write the names given to bleeding into the four body cavities. Underneath, write which pattern points to a platelet or small vessel problem and which points to a coagulation factor deficiency, and explain each from what platelets do and what fibrin does at an injured vessel. Finish by writing the color changes of a bruise in order, naming the pigment responsible for each stage, and what a large resorbing hematoma can do to the serum bilirubin.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C05-BD2

Volume, rate, and site in acute blood loss

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C05 Activity: NPLEX-P1-CARD-18-08-C05-BD2 Date: _____

A man weighing 70 kg is brought in 30 minutes after losing about 1.5 L of blood from a stab wound. He is pale and tachycardic, and his hematocrit is 44%. Calculate his blood volume and the percentage he has lost, then write what percentage of blood volume is tolerated with little change, what percentage brings on hypovolemic shock, and what percentage is immediately life threatening. Explain why his hematocrit is normal now, what two processes will lower it over the next day, and why an early normal hematocrit does not exclude severe blood loss. Then write what the baroreceptor reflex is doing to his heart rate, his arterioles, his veins, and his pulse pressure. Finish by writing why a 50 mL bleed inside the skull, a 150 mL bleed into the pericardium, and a 2 L bleed into the retroperitoneum each do what they do, and why slow recurrent external bleeding causes an anemia that bleeding into a body cavity does not.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C06-BD1

The mechanisms of edema and the fluid each produces

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-BD1 Date: _____

Write the one sentence definition of edema, then write what normally keeps the interstitium dry, naming the forces at the capillary and the structure that returns the excess. Build a five row table of the mechanisms of edema with columns for the mechanism, two or three examples, and the kind of fluid produced. Underneath, write the features that separate a transudate from an exudate: protein content, cell content, specific gravity, clarity, and the state of the vessel wall in each. Name the mediators that open the wall in an exudate. Finish by writing the three pleural fluid criteria that mark a fluid as an exudate, with the number for each.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C06-BD2

Four swollen patients, four mechanisms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-BD2 Date: _____

Four patients have swelling. The first has nephrotic syndrome with heavy albumin loss in the urine. The second has long standing heart failure. The third has had an axillary lymph node dissection for breast cancer and has a swollen arm. The fourth has untreated hypothyroidism. For each patient write the mechanism, the protein content of the retained fluid, whether the swelling pits, and where on the body it appears first, and explain each answer. For the third patient explain why the swelling is soft at first and hard years later, and name what the skin and subcutaneous tissue become. For the fourth explain why the swelling does not pit even though it is not lymphedema. Finish by writing why ordinary edema collects in the legs when standing and over the sacrum when lying.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C06-RT01

Rebuild the edema cycle in heart failure

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C06 Activity: NPLEX-P1-CARD-18-08-C06-RT01 Date: _____

Rebuild the six step cycle that runs from a fall in cardiac output to a further overloaded heart, one step per line and in order, then draw it as a loop with arrows. Beside the renal step name the hormones involved and what each one retains. Beside the step where edema forms, name the Starling force that has changed and write whether the fluid is a transudate or an exudate. Then write in two sentences why a response meant to restore perfusion makes the problem worse, and mark on your loop the one point at which lowering venous pressure would interrupt it.

Rebuild from memory: Edema in heart failure

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C07-BD1

Build the hemodynamic table of shock

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C07 Activity: NPLEX-P1-CARD-18-08-C07-BD1 Date: _____

Write the one sentence definition of shock and the equation that relates mean arterial pressure to cardiac output and systemic vascular resistance, then write what cardiac output itself depends on. Name the three measurements used to identify the type of shock and write what the third one approximates. Now build a table from memory with one row each for hypovolemic, cardiogenic, obstructive, septic, anaphylactic, and neurogenic shock, and columns for the primary problem, cardiac output, systemic vascular resistance, pulmonary capillary wedge pressure, and the skin. Give at least two causes in the primary problem cell of each row. Finish by writing which rows show a reflex change in resistance and which show resistance as the primary defect, and write the normal range for each of the three measurements.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C07-BD2

Read four hemodynamic profiles

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C07 Activity: NPLEX-P1-CARD-18-08-C07-BD2 Date: _____

Four patients are measured. Patient A has a cardiac output of 2.5 L/min, systemic vascular resistance of 1,800 dyn.s/cm5, and a wedge pressure of 26 mm Hg. Patient B has an output of 2.8 L/min, resistance of 1,700, and a wedge pressure of 4 mm Hg. Patient C has an output of 8.5 L/min, resistance of 450, and a wedge pressure of 8 mm Hg. Patient D has an output of 2.4 L/min, resistance of 1,900, a wedge pressure of 7 mm Hg, and a central venous pressure of 20 mm Hg. For each patient name the type of shock and write, value by value, which measurement is the primary problem and which is the reflex response. For patient D explain how the central venous pressure and the wedge pressure can move in opposite directions and name the two causes of shock that do this. Then write what tamponade does to the diastolic pressures of all four chambers and why. Finish with the one patient among these four whose mixed venous oxygen saturation would be normal or high, and explain why tissues are still hypoxic.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-18-08-C07-CP01

Cardiogenic against hypovolemic against septic shock

Fill in every cell from memory, then check it against the notes and correct what you missed.

Feature	Cardiogenic shock	Hypovolemic shock	Septic shock early
Primary problem and two causes			
Cardiac output and whether it is the primary change or a response			
Systemic vascular resistance and whether it is primary or reflex			
Pulmonary capillary wedge pressure and what it says about the left ventricle			
Skin temperature and appearance			
Mixed venous oxygen saturation			
Why the tissues are underperfused in this type			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-18-08-C07-PR01

Predict the profile after a spinal cord injury at T4

A man has a complete spinal cord injury at the fourth thoracic level and becomes hypotensive within minutes, with no evidence of bleeding. Predict each item below, give the direction of the change, and write the mechanism in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to sympathetic outflow to the arterioles and veins below the injury		
What happens to systemic vascular resistance and whether that change is primary or reflex		
What happens to venous return and to the pulmonary capillary wedge pressure		
What happens to cardiac output		
What happens to heart rate, and how this differs from every other type of shock with a low cardiac output		
What the skin feels and looks like, and why it differs from the skin in hypovolemic shock		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-18-08-C08-BD1

The three stages of shock and the organs that fail

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-BD1 Date: _____

Write out the three stages of shock in order. For each stage name what the body is doing or failing to do, the mechanisms and hormones involved, and the findings in the patient, including the blood pressure, the urine output, the mental state, and the skin. For the middle stage write the acid base disturbance by name, the metabolic pathway that causes it, the substance that accumulates, and its normal upper value. Then build a table of multiorgan dysfunction with one row each for kidney, lung, blood, liver, and adrenal, and columns for the lesion and the mechanism with its findings. Finish by writing the one sentence rule that explains why each organ fails where it does.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-18-08-C08-BD2

From one gram negative bacterium to failure in five organs

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-BD2 Date: _____

A woman with a gram negative urinary infection becomes hypotensive with warm flushed skin, a rising creatinine, hypoxemia, a falling platelet count, and a rising D-dimer. Starting from the bacterial molecule itself, write the chain that explains each of her findings: name the molecule and its toxic portion, the receptors and binding proteins it uses, the transcription factor activated, the two cytokines released first, and the enzyme and gas that lower her vascular resistance. Then take each finding in turn and write which step in that chain produced it. Explain why her coagulation tests are abnormal in both directions at once, naming the three changes cytokines make to the hemostatic balance. Explain why her kidney injury appears in the outer medulla rather than evenly through the kidney, and why her hypoxemia occurs with a normal left atrial pressure. Finish by writing why her shock would continue to worsen even if the infection were cleared at this point, and name the stage at which restoring her circulation would no longer help.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-18-08-C08-RT01

Rebuild the endotoxin sequence and the vicious cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-18-H Hemodynamic Disorders: Thrombosis, Embolism, Infarction, Hemorrhage, Edema, and Shock Competency: NPLEX-P1-CARD-18-08-C08 Activity: NPLEX-P1-CARD-18-08-C08-RT01 Date: _____

Rebuild the six step sequence that runs from endotoxin binding its receptors to multiorgan dysfunction, one step per line and in order. Beside the second step name the transcription factor and the two cytokines. Beside the third step name the enzyme, the gas it makes, and the second messenger through which that gas relaxes vascular smooth muscle. Beside the fourth step write what happens to tissue factor and to protein C activation and what that combination produces in the small vessels. Underneath, rebuild the six step positive feedback cycle of progressive shock as a loop, and mark the two points at which falling flow causes a further fall in flow.

Rebuild from memory: From endotoxin to septic shock, and the positive feedback cycle of progressive shock

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-18-08-C08-CA01

Hypotension and bleeding gums three days after a fracture

A 58-year-old woman is in hospital on the third day after surgical fixation of a femoral fracture. She becomes confused and her urine output falls to 15 mL/h. Temperature is 38.9 C (102.0 F), pulse is 124/min, respirations are 28/min, and blood pressure is 82/44 mm Hg. Her hands and feet are warm and the skin is flushed. Urine obtained before surgery grew Escherichia coli. Laboratory studies show a platelet count of 48,000/mm3, fibrinogen of 95 mg/dL, prolonged prothrombin and partial thromboplastin times, a D-dimer well above the reference range, schistocytes on the smear, serum lactate of 5.8 mmol/L, and creatinine rising from 0.9 to 2.6 mg/dL. The urine sediment contains muddy brown granular casts. Arterial blood shows hypoxemia, and the chest radiograph shows bilateral infiltrates. Blood is oozing from her gums and from both intravenous sites.

1. Recognize. Name the type of shock, the stage it has reached, and each of the three organ lesions her results point to.

2. Explain. Starting from the bacterial molecule, write the chain of mediators that produced her low vascular resistance, her warm flushed skin, and her capillary leak.

3. Differentiate. Write how her hemodynamic profile and her skin differ from the profile and skin of cardiogenic and hypovolemic shock, and name the measurement that would separate the three.

4. Investigate. Write which measurements you would obtain to confirm each of the three organ lesions, and the direction or appearance each result would take.

5. Decide. State which single process explains both the microthrombi and the bleeding, and state whether her shock at this point would respond to restoring her circulation.

6. Justify. Justify that answer from what cytokines do to tissue factor, thrombomodulin, and plasminogen activator inhibitor-1, and from what defines the stage of shock you named.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-18-08-C03-Q01

Sudden collapse after a flight

A 64-year-old woman collapses in an airport after a 12-hour flight. On arrival she is hypotensive with distended neck veins and clear lung fields. Arterial blood gas analysis shows a low partial pressure of oxygen and a low partial pressure of carbon dioxide. She dies within the hour. At autopsy a large mass of layered thrombus sits astride the bifurcation of the main pulmonary artery, and the right ventricle is dilated. Which of the following best explains the low arterial carbon dioxide in this patient?

A. Hyperventilation driven by hypoxemia and stimulated receptors, with ventilation continuing in lung that is no longer perfused

B. Reduced carbon dioxide production by hypoperfused peripheral tissues

C. Loss of alveolar surface area from collapse of the affected lung segments

D. Metabolic alkalosis from catecholamine release during the collapse

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C07-Q02

A profile with a high filling pressure

A 71-year-old man is hypotensive 6 hours after the onset of chest pain. His skin is cool and mottled. Measurements show a cardiac output of 2.4 L/min, a systemic vascular resistance of 1,900 dyn.s/cm5, and a pulmonary capillary wedge pressure of 28 mm Hg. Crackles are heard over both lung fields. Which of the following best accounts for the combination of these three measurements?

A. Loss of circulating volume with reflex arteriolar constriction

B. Reduced sympathetic tone with venous pooling below the lesion

C. Failure of the left ventricle to empty, with reflex arteriolar constriction

D. Cytokine-driven vasodilation with capillary leak

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C02-Q03

A long clotting time with thrombosis

A 29-year-old woman has a deep vein thrombosis of the left leg and a history of two second-trimester pregnancy losses. Her partial thromboplastin time is prolonged and does not correct fully when her plasma is mixed with normal plasma. A nontreponemal screening test for syphilis is positive, and confirmatory treponemal testing is negative. Which of the following mechanisms best explains her thrombosis?

A. Resistance of factor Va to cleavage by activated protein C

B. Antibodies against beta2-glycoprotein I that activate platelets, endothelium, and complement

C. Deficiency of antithrombin, so thrombin and factor Xa are not inactivated

D. Increased plasma prothrombin from a mutation in the untranslated region of its gene

Why did you pick it?

Board-style question NPLEX-P1-CARD-18-08-C04-Q04

A cavity where an infarct had been

An 80-year-old man had a sudden loss of speech and right-sided weakness 8 months ago. He dies of unrelated causes. At autopsy the left frontal and temporal cortex contains a fluid-filled cavity bordered by reactive astrocytes, with no scar tissue. A similar sized old infarct in his left kidney is a firm pale depressed scar. Which of the following best explains why these two infarcts healed so differently?

A. The kidney infarct was white while the brain infarct was red, and blood in the dead tissue prevents scarring

B. The brain was reperfused when the embolus lysed, while the renal artery remained occluded

C. Neurons tolerate hypoxia for only a few minutes while renal tubular cells tolerate it for hours

D. Brain tissue has little collagen framework and is lipid rich, so lysosomal enzymes liquefy it and microglia clear it away

Why did you pick it?